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Dynamical Systems III

With 81 Figures



Springer-Verlag
Berlin Heidelberg New York
London Paris Tokyo

Proposition 16. If $H''_{0QQ}|_{Q=f} \neq 0$, then equations (53) admit a unique solution given by formal power series in ε :

$$\begin{aligned} p &= p_0(t) + \varepsilon p_1(t) + \dots, & q &= q_0(t) + \varepsilon q_1(t) + \dots, \\ P &= \varepsilon P_1(t) + \dots, & Q &= f(p_0(t), q_0(t)) + \varepsilon Q_1(t) + \dots, \end{aligned}$$

where $p_0(t), q_0(t)$ is some prescribed solution of equations (52).

Unfortunately, these series do not always converge. In the case where function H_0 does not depend on Q equations (53) are no longer singular: instead of the momentum P we must take the new variable P/ε . Then the solutions of these equations can be expressed as convergent power series, and the initial conditions $Q(0)$ and $\dot{Q}(0)$ may be arbitrary. This is precisely the case in the "restricted" n -body problem, when one of the masses tends to zero.

Thus, Dirac's mechanics may be viewed as a mechanics of small masses. On the contrary, vakonomic mechanics is convenient for describing the dynamics of large masses.

The assertions of this subsection may serve as motivation for our theoretical constructions concerning the dynamics of constrained mechanical systems.

Chapter 2 The n -Body Problem

§1. The Two-Body Problem

1.1. Orbits. Suppose that two points (r_1, m_1) and (r_2, m_2) mutually interact with potential energy $U(|r_1 - r_2|)$, so that the equations of motion have the form

$$m_1 \ddot{r}_1 = -\frac{\partial U}{\partial r_1}, \quad m_2 \ddot{r}_2 = -\frac{\partial U}{\partial r_2}.$$

Proposition 1. The relative position vector $r = r_1 - r_2$ in the two-body problem changes according to the same law as the position vector of a point of mass $m = m_1 m_2 / (m_1 + m_2)$ moving in a central force field with potential $U(|r|)$.

If

$$\xi = \frac{m_1 r_1 + m_2 r_2}{m_1 + m_2}$$

is the barycenter of the two points m_1 and m_2 , then obviously

$$r_1 = \xi + \frac{m_2}{m_1 + m_2} r, \quad r_2 = \xi - \frac{m_1}{m_1 + m_2} r.$$

It follows from this formulas that in a barycentric coordinate system the trajectories of the two point masses are similar planar curves (with similarity ratio m_1/m_2). Therefore, the problem reduces to the investigation of the single equation

$$m \ddot{r} = -\frac{\partial U}{\partial r}, \quad r \in R^3.$$

Let x, y be Cartesian coordinates in the plane of the orbit. Then $K_z = m(x\dot{y} - y\dot{x}) = \text{const}$. In polar coordinates $x = r \cos \varphi$, $y = r \sin \varphi$, we have $K_z = m r^2 \dot{\varphi}$. Consequently, $r^2 \dot{\varphi} = c = \text{const}$. If $c = 0$, then $\varphi = \text{const}$ (the point moves along a straight line). We shall assume that $c \neq 0$. In this case φ is a monotonic function of t , and hence the inverse function $t = t(\varphi)$ exists. As the point m moves, its position vector sweeps out a curvilinear sector of area

$$S(t) = \frac{1}{2} \int_{\varphi(0)}^{\varphi(t)} r^2 d\varphi = \frac{1}{2} \int_0^t r^2 \dot{\varphi} dt = \frac{ct}{2}.$$

Therefore, $\dot{S} = c/2 = \text{const}$ (i.e., the "sectorial velocity" is constant). This fact is known as the *area integral* or as *Kepler's second law*; c is called the area constant.

Proposition 2 (Newton). For a fixed value of the area integral c

$$m\ddot{r} = -\frac{\partial U_c}{\partial r} \quad \text{with} \quad U_c = U + \frac{mc^2}{2r^2} \quad (r > 0). \quad (1)$$

This equation describes the motion of a point of mass m on the half line $R^+ = \{r > 0\}$ under the action of the conservative force with potential U_c . It is integrable by quadratures using the energy integral

$$\frac{m\dot{r}^2}{2} + U_c = h.$$

The function U_c is called the *reduced (effective, or amended) potential*.

The energy and area integrals can be used to find the equation of orbits without solving equation (1). In fact, since $\dot{r} = \sqrt{2(h - U_c)/m}$ and $r^2\dot{\varphi} = c$, it follows that

$$\frac{dr}{d\varphi} = \frac{dr}{dt} \frac{dt}{d\varphi} = \frac{r^2}{c} \sqrt{\frac{2(h - U_c)}{m}}.$$

Integrating this equation we get

$$\varphi = \int \frac{c \, dr}{r^2 \sqrt{\frac{2(h - U_c)}{m}}}.$$

In calculating orbits it is sometimes useful to keep in mind the following proposition:

Proposition 3 (Clairaut). Let $\rho = 1/r$ and write the equation of the orbit as $\rho = \rho(\varphi)$. Then

$$m \frac{d^2 \rho}{d\varphi^2} = -\frac{1}{c^2} \frac{d}{d\rho} U_c \left(\frac{1}{\rho} \right).$$

For fixed values h and c the orbit lies in the domain

$$B_{c,h} = \left\{ (r, \varphi) \in R^2 : U + \frac{mc^2}{2r^2} \leq h \right\},$$

which is a union of several annuli. Now suppose that h is not a critical value of the reduced potential U_c and that $B_{c,h}$ coincides with the annulus $0 < r_1 \leq r \leq r_2 < \infty$.

We show that in this cases $r(t)$ is a periodic function of time, and that

$$\min r(t) = r_1, \quad \max r(t) = r_2.$$

To see this, let

$$u = \frac{\pi}{\tau} \int_{r_1}^r \frac{dx}{\sqrt{\frac{2}{m}(h - U_c(x))}}, \quad \tau = \int_{r_1}^{r_2} \frac{dx}{\sqrt{\frac{2}{m}(h - U_c(x))}}.$$

Then obviously $r(u)$ is a 2π -periodic function of u , and $\dot{u} = \pi/\tau = \text{const}$. The period of the function $r(\cdot)$ is clearly 2τ .

Angle φ varies monotonically (provided, of course, that $c \neq 0$). The points on the orbit lying closest (farthest) to the center are called *pericenters* (respectively, *apocenters*). The orbit is symmetric with respect to the lines passing through the point $r=0$ and its pericenters (apocenters). The angle Φ between the lines joining the center with two adjacent apocenters (or pericenters) is called the *apsidal angle*. The orbit is invariant under rotation through angle Φ . If the apsidal angle

$$\Phi = 2 \int_{r_1}^{r_2} \frac{c \, dr}{r^2 \sqrt{\frac{2}{m}(h - U_c)}}$$

is commensurable with π , the orbit is closed; otherwise it densely fills the annulus $B_{c,h}$. If $r_2 = \infty$ the orbit is unbounded.

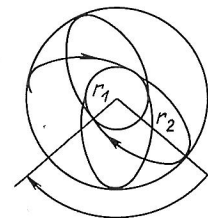


Fig. 8. Orbit in a central field

The motion of the point on a circle $r=r_0$ is called a *relative equilibrium*. It is clear that such a motions are uniform and that the values r_0 are the critical points of the reduced potential U_c . If at the point $r=r_0$ function U_c has a local minimum, then the corresponding circular motion is orbitally stable.

Theorem 1 (J.L.F. Bertrand). Suppose that for some $c \neq 0$ there exists a stable relative equilibrium and that the potential U_c is analytic for $r > 0$. If all orbits sufficiently close to the given circular orbit are closed, then either $U = \gamma r^2$ or $U = -\gamma/r$ (where $\gamma > 0$).

In the first case the system is a harmonic oscillator and its orbits are ellipses centered at the point $r=0$. The second case is that of gravitational attraction. The problem of the motion of a point in a conservative field with potential $U = -\gamma/r$ is usually called *Kepler's problem*.

The reduced potential in Kepler's problem is

$$U_c = \frac{c^2}{2r^2} - \frac{\gamma}{r}.$$

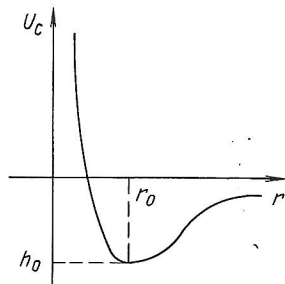


Fig. 9. Reduced potential of Kepler's problem

According to *Clairaut's equation* (Proposition 3), $\frac{d^2 \rho}{d\varphi^2} = -\rho + \frac{\gamma}{c^2}$. This linear nonhomogeneous equation is readily solved:

$$\rho = A \cos(\varphi - \varphi_0) + \frac{\gamma}{c^2} = \frac{1}{p} (1 + e \cos(\varphi - \varphi_0)), \quad (2)$$

where e, φ_0 are constants, and $p = c^2/\gamma > 0$. Hence

$$r = \frac{p}{1 + e \cos(\varphi - \varphi_0)},$$

so that the orbits of Kepler's problem are conics with one focus at the center of attraction (this is *Kepler's first law*).

For fixed $c \neq 0$ there exists a unique relative equilibrium $r_0 = c^2/\gamma$. Its energy $h_0 = -\gamma^2/2c^2$ is minimal. Using the simple formula

$$v^2 = \dot{r}^2 + r^2 \dot{\varphi}^2 = c^2(\rho'^2 + \rho'^2), \quad \rho' = \frac{d\rho}{d\varphi},$$

we can rewrite the energy integral as

$$\frac{c^2}{2} (\rho'^2 + \rho^2) - \gamma \rho = h.$$

On substituting in this formula the equation of the orbit (2) we obtain the expression of the eccentricity: $e = \sqrt{1 + 2c^2 h/\gamma^2}$. Since $h \geq h_0 = -\gamma^2/2c^2$, the eccentricity assumes only real values.

If $h = h_0$, then $e = 0$ and the orbit is circular. If $h_0 < h < 0$, then $0 < e < 1$ and the orbit is an ellipse. If $h = 0$, then $e = 1$ and the orbit is a parabola. Finally, if $h > 0$, then $e > 1$ and the point moves along one of the branches of a hyperbola.

Figure 10 shows the *bifurcation set* Σ in the plane of the parameters c, h . It consists of the curve $h = -\gamma^2/2c^2$ and the two coordinate axes $c = 0$ and $h = 0$. At the points of Σ the topological type of the region of possible motions $B_{c,h}$ changes (in Fig. 10 the regions $B_{c,h}$ are shaded).

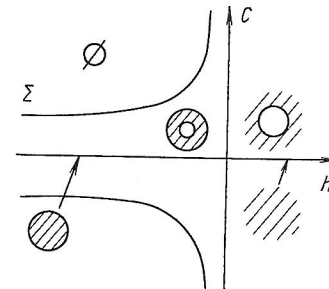


Fig. 10

In the case of the harmonic oscillator the period of revolution (orbital period) does not depend on the initial configuration. In Kepler's problem the picture is different. Thus, for elliptic motions "*Kepler's third law*" holds: $a^3/T^2 = \gamma/4\pi^2 = \text{const}$, where a is the major semiaxis of the ellipse and T is the period of revolution. Since $a = \frac{p^2}{1-e^2} = \frac{\gamma}{2|h|}$, T depends only on the constant.

1.2. Anomalies. To solve Kepler's problem completely we must still determine the law governing the motion on the already known orbits. To this end, we take the coordinate axes x and y along the principal axes of the conic representing the given orbit. The equation of this conic can be written in the following parametric form:

$$\begin{aligned} x &= a(\cos u - e), & y &= a\sqrt{1-e^2} \sin u & (0 \leq e < 1), & \text{if } h < 0, \\ x &= a(\cosh u - e), & y &= a\sqrt{e^2-1} \sinh u & (e > 1), & \text{if } h > 0, \\ x &= \frac{1}{2}(p - u^2), & y &= \sqrt{p}u, & & \text{if } h = 0. \end{aligned} \quad (3)$$

In astronomy the auxiliary variable u is known as the *eccentric anomaly*, and the angle φ made by the line joining the center to the pericenter (the x -axis) and the position vector of the point – as the *true anomaly*. These anomalies are related by

$$\tan \frac{\varphi}{2} = \begin{cases} \sqrt{\frac{1+e}{1-e}} \tan \frac{u}{2}, & \text{if } h < 0, \\ \sqrt{\frac{e+1}{e-1}} \tanh u, & \text{if } h > 0, \\ \frac{u}{p}, & \text{if } h = 0. \end{cases}$$

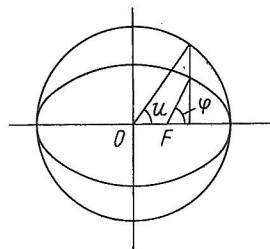


Fig. 11

Substituting expressions (3) in the area integral $x\dot{y} - y\dot{x} = c$ and integrating we obtain the relation between time and eccentric anomaly:

$$u - e \sin u = n(t - t_0), \quad n = \frac{\sqrt{\gamma}}{p^{3/2}}, \quad \text{if } h < 0,$$

$$u - e \sinh u = n(t - t_0), \quad n = -\frac{\sqrt{\gamma}}{p^{3/2}}, \quad \text{if } h > 0,$$

$$u + \frac{u^3}{3p} = n(t - t_0), \quad n = \frac{2\sqrt{\gamma}}{p}, \quad \text{if } h = 0.$$

Here t_0 represents the time of passage at apocenter. These equations (or at least the first of them) are called *Kepler's equations*. The linear function $\zeta = n(t - t_0)$ is usually referred to as the *mean anomaly*.

Thus, in the elliptic case of Kepler's problem we must solve the transcendental equation of Kepler

$$u - e \sin u = \zeta.$$

It is clear that for $0 \leq e < 1$ this equation has an analytic solution $u(e, \zeta)$ such that the difference $u(e, \zeta) - \zeta$ is 2π -periodic in the mean anomaly ζ . To represent the function $u(e, \zeta)$ in a form suitable for computations we may proceed in two ways:

1) expand the difference $u - \zeta$ for fixed values of e in a Fourier series in variable ζ with coefficients depending on e , or

2) attempt to write $u(e, \zeta)$ as a power series in the eccentricity e with coefficients depending on ζ .

The first approach gives

$$u = \zeta + 2 \sum_{m=1}^{\infty} \frac{J_m(me)}{m} \sin m\zeta, \quad (4)$$

where

$$J_m(z) = \frac{1}{2\pi} \int_0^{2\pi} \cos(mx - z \sin x) dx = \sum_{k=0}^{\infty} \frac{(-1)^k (z/2)^{m+2k}}{k!(m+k)!}, \quad m=0, 1, \dots,$$

are the *Bessel function* of order m . These functions, "though usually associated with the name of Bessel, have been used extensively, precisely in this connection (which is that of Bessel), and more than half a century prior to Bessel, by Lagrange and others" (A. Winter, [42]).

The proof of expansion (4) is a straightforward calculation:

$$\begin{aligned} \frac{du}{d\zeta} &= \frac{1}{1 - e \cos u} = \frac{1}{2\pi} \int_0^{2\pi} \frac{d\zeta}{1 - e \cos u} + \sum_{m=1}^{\infty} \frac{\cos m\zeta}{\pi} \int_0^{2\pi} \frac{\cos m\zeta d\zeta}{1 - e \cos u} = \\ &= \frac{1}{2\pi} \int_0^{2\pi} du + \sum_{m=1}^{\infty} \frac{\cos m\zeta}{\pi} \int_0^{2\pi} \cos [m(u - e \sin u)] du = \\ &= 1 + 2 \sum_{m=1}^{\infty} J_m(me) \cos m\zeta, \end{aligned}$$

and now it remains to integrate this formula with respect to ζ .

In the second approach we get the expansion

$$u(e, \zeta) = \sum_{m=0}^{\infty} c_m(\zeta) \frac{e^m}{m!}, \quad (5)$$

where

$$c_m(\zeta) = \left. \frac{\partial^m u(e, \zeta)}{\partial e^m} \right|_{e=0}.$$

Using Lagrange's well-known formula for the local inversion of holomorphic functions (which he discovered exactly in his attempt to solve Kepler's problem), we obtain the expressions for the coefficients of series (5):

$$c_0(\zeta) = \zeta; \quad c_m(\zeta) = \frac{d^{m-1}}{d\zeta^{m-1}} \sin^m \zeta, \quad m \geq 1.$$

The functions $c_m(\zeta)$ are trigonometric polynomials in the mean anomaly ζ . Rearranging the terms of series (5) one can obtain expansion (4). As a matter of fact, this is the way in which Lagrange arrived at formula (4).

By the implicit function theorem (and in view of the periodicity of the function $u(e, \zeta) - \zeta$) series (5) converge on the entire real axis $\zeta \in \mathbb{R}$ for e small enough. A careful analysis of expansion (4) shows that Lagrange's series converges for $e \leq 0.6627434 \dots$ ¹⁵.

1.3. Collisions and Regularization. Thus far we have assumed that the area constant $c \neq 0$. Now let $c = 0$. In this case the point moves along a straight line, and we may assume that the latter is the x -axis. If at some instant the velocity $\dot{x}$ is directed toward the center of attraction, then $x(t) \rightarrow 0$ and

¹⁵ "In fact, a principal impetus for Cauchy's discoveries in complex function theory was his desire to find a satisfactory treatment for Lagrange's series." (A. Wintner [42]).

$\dot{x}(t) \rightarrow \infty$ as t approaches a certain value t_0 . Therefore, at time $t = t_0$ a collision of the two bodies occurs. It is obvious that for $c=0$ the function $x(t)$, $t \in \mathbb{R}$, has necessarily a singularity of the indicated type.

We show that the eccentric anomaly u is a *regularizing variable*, which resolves the singularity of the analytic function $x(t)$. Since $c=0$, $e=1$ in the elliptic and hyperbolic cases, and $p=0$ in the parabolic case. Consequently, formulas (3) take respectively the form

$$x = a(\cos u - 1), \quad x = a(\cosh u - 1), \quad x = -\frac{u^2}{2}. \quad (6)$$

It follows from these formulas that if $h < 0$ a collision occurs for $u = 2\pi k$, $k \in \mathbb{Z}$, whereas if $h \geq 0$ it occurs only for $u = 0$. Actually, in the elliptic case as well it suffices to consider the value $u = 0$.

To simplify the discussion we set $t_0 = 0$. From Kepler's equation (for $e = 1$) it readily follows that in the neighborhood of the point $u = 0$ we can write

$$t = u^3 f(u),$$

where f is a function analytic in the neighborhood of zero, and $f(0) \neq 0$. From (6) we derive an analogous representation

$$x = u^2 g(u)$$

with an analytic function g , $g(0) \neq 0$. Upon eliminating the eccentric anomaly u from the last two formulas we get the Puiseux expansion

$$x(t) = (\sqrt[3]{t})^2 \sum_{n=0}^{\infty} c_n (\sqrt[3]{t})^n.$$

Here the coefficients c_n with n odd obviously vanish, and $c_0 \neq 0$. Therefore, $x(t)$ is an even function of time, i.e., the moving point is reflected back from the center of attraction after collision. If x and t are regarded as complex variables, then $t=0$ is an algebraic branching point of the analytic function $x(t)$. At the collision point $t=0$ the three sheets of the Riemann surface of $x(t)$ merge; also, the real values of $x(t)$ for $t > 0$ and $t < 0$ lie on the same sheet. Hence, the function $x(t)$ admits a unique real analytic continuation.¹⁶

In conclusion we mention a result of K. Bohlin concerning the regularization of the two-body problem in the general elliptic case ($h < 0$).

Introducing the complex variable $z = x + iy$ we rewrite the equation of Kepler's problem and the energy integral as

$$\ddot{z} = -\frac{\gamma z}{|z|^3}$$

and

$$\frac{|\dot{z}|^2}{2} = \frac{\gamma}{|z|} + h, \quad (7)$$

¹⁶ The regularization of collisions in the two-body problem goes back to Euler.

respectively. Next, we change the independent variable z to w and the time variable t to τ according to the rules

$$z = w^2, \quad t' = \frac{dt}{d\tau} = 4|w^2| = 4|z|. \quad (8)$$

In the new variables w, τ , equation (7) becomes

$$\frac{|w'|^2}{2} = 4\gamma + 4h|w^2|. \quad (9)$$

This further becomes $w'' + 8|h|w = 0$, i.e., the equation describing the vibrations of a harmonic oscillator. Thus, the nonlinear mapping (8) takes the orbits of Kepler's problem with energy constant $h < 0$ into orbits of the harmonic oscillator lying on the energy level (9). This conclusion is an interesting supplement to Bertrand's theorem.

The regularizing variable τ depends linearly on the eccentric anomaly u . In fact, since

$$|z| = r = a(1 - e \cos u) \quad \text{and} \quad nt = u - e \sin u,$$

it follows that

$$\frac{du}{d\tau} = \frac{n}{1 - e \cos u} = \frac{na}{r},$$

whence $u = 4na\tau$.

1.4. Geometry of the Kepler Problem. J. Moser observed that by suitably changing the time variable one can transform the phase flow of Kepler's problem into the geodesic flow on a surface of constant curvature. In our presentation of this result we follow Yu. S. Osipov (Usp. Mat. Nauk **27**, No. 2, 161 (1972)).

Lemma 1. Let $x(t)$ be a solution of the Hamiltonian system with Hamiltonian $H(x)$ which lies on the level surface $H=0$. Change the time variable t to τ along this trajectory according to the formula $d\tau/dt = 1/G(x(t)) \neq 0$. Then $x(\tau) = x(t(\tau))$ is a solution of the Hamiltonian (with respect to the same symplectic structure) system with Hamiltonian $\tilde{H} = HG$. If $G = 2(H + \alpha)$ with $\alpha = \text{const}$, one can also put $\tilde{H} = (H + \alpha)^2$.

Let us write the Hamiltonian of Kepler's problem in the notations of §3.1: $H = |p|^2/2 - \gamma/|z|$, with $p = \dot{z}$. Now perform the change of time $\tau = |z|^{-1}$ on the submanifold $H=h$ (cf. formula (8)). According to Lemma 1, this leads to the new Hamiltonian $|z|(H-h) = |z|(|p|^2 - 2h)/2 - \gamma$. Then again change the time τ to τ by $d'\tau/d\tau = 2|z|((H-h) + \gamma)$ on the same level surface $H=h$. In the end we get the Hamiltonian system with Hamiltonian

$$\tilde{H} = |z|^2(p^2 - 2h)/4.$$

Finally, apply Legendre's transformation, regarding p as a coordinate and z as the canonically conjugate momentum. We thus obtain a natural La-

grangian system with the Lagrangian

$$L = \frac{|p'|^2}{(2h - |p|^2)^2}. \quad (10)$$

This (length) function defines a Riemann metric of constant Gaussian curvature (positive for $h < 0$ and negative for $h > 0$). In the case $h < 0$ the geodesics of the metric (10) (which is defined for all $p \in R^2$) are images of great circles on the sphere under stereographic projection, whereas in the case $h > 0$ (in which the metric is defined in the disc $|p|^2 < 2h$) the geodesics are the straight lines of Lobachevsky's geometry (in the Poincaré model).

Remark (A.B. Givental'). Suppose $\{(\dot{x}, \dot{y})\}$ is the configuration plane of the Kepler problem with the Lagrangian $L = (\dot{x}^2 + \dot{y}^2)/2 + 1/\sqrt{x^2 + y^2}$. In the space $\{(x, y, z)\}$ consider the right circular cone $z^2 = x^2 + y^2$ and the family of inscribed paraboloids of revolution $z = (x^2 + y^2)/4\alpha + \alpha$, where α is a parameter. By "projection" we mean the projection of the space $\{(x, y, z)\}$ onto the plane $\{(x, y)\}$ parallel to the z -axis. Then one can show that

1) the trajectories of Kepler's problem are projections of plane sections of the cone (in particular, the vertex of the cone is a focus of these projections);

2) trajectories with the same value of the total energy are projections of sections of the cone by planes tangent to one and the same paraboloid;

3) trajectories with the same value of the angular momentum are projections of sections of the cone by planes passing through one and the same point of the z -axis.

§2. Collisions and Regularization

2.1. Necessary Conditions for Stability. We now turn to the general n -body problem in which n point masses $(m_1, r_1), \dots, (m_n, r_n)$ attract one another according to the law of universal attraction. The kinetic energy of this system is

$$T = \frac{1}{2} \sum m_i \dot{r}_i^2,$$

and the force function

$$V = \sum_{j < k} \frac{m_j m_k}{|r_{jk}|}, \quad r_{jk} = r_j - r_k,$$

is everywhere positive. Consider an inertial reference frame with origin at the center of mass and let r_i be the position vectors of the points in this new frame. The equations of the n -body problem are Lagrange's equations with the Lagrangian $L = T + V$.

We say that the motion $r_k(t)$ ($1 \leq k \leq n$) is *stable* if the following two conditions are fulfilled:

- a) $r_{ij}(t) \neq 0$ for all values of t and all $i \neq j$ (absence of collisions).
- 2) $|r_{ij}(t)| \leq c$ ($c = \text{const}$).

Theorem 2 (C. Jacobi). *If a motion is stable, then its total energy $h = T - V$ is negative.*

◁ We apply Lagrange's formula

$$\dot{I} = 2V + 4h, \quad (11)$$

where $I = \sum m_i r_i^2$ is the polar moment of inertia. If $h \geq 0$, then the function $I(t)$, $t \in R$, is upward convex and hence cannot be simultaneously bounded from below and from above. To complete the proof it remains to use Lagrange's identity:

$$I \sum m_i = \sum_{j < k} m_j m_k r_{jk}^2 + (\sum m_i r_i)^2. \triangleright$$

Under the additional assumption that mutual distances are bounded from below ($|r_{ij}(t)| \geq c > 0$), it follows from the energy integral and Lagrange's formula (11) that along a stable motion the mean values

$$\lim_{s \rightarrow \infty} \frac{1}{s} \int_0^s V(t) dt \quad \text{and} \quad \lim_{s \rightarrow \infty} \frac{1}{s} \int_0^s 2T(t) dt$$

exist and are equal to $-2h > 0$.

The necessary condition for stability $h < 0$ is not sufficient for $n > 2$.

2.2. Simultaneous Collisions. If the position vectors $r_i(t)$ of all points tend to the same limit r_0 as $t \rightarrow t_0$, then we say that a simultaneous (or general) collision occurs at time t_0 . The point r_0 must obviously coincide with the center of mass, i.e., $r_0 = 0$. A simultaneous collision takes place if and only if the polar moment of inertia $I(t)$ tends to zero as $t \rightarrow t_0$.

Theorem 3. *If $I(t) \rightarrow 0$ as $t \rightarrow t_0$, then the constant angular momentum vector*

$$K = \sum m_i (r_i \times \dot{r}_i)$$

is equal to zero.

For $n = 3$ this fact was already known to Weierstrass.

◁ Since $V(t) \rightarrow +\infty$ as $t \rightarrow t_0$, it follows from the equation $\dot{I} = 2V + 4h$ that $\dot{I}(t) > 0$ for t close to t_0 . Hence, prior to collision $I(t)$ decreases monotonically.

We now use the inequality $K^2 \leq 2IT$ (given in §1.1) which, in view of Lagrange's formula, is equivalent to

$$\dot{I} \geq \frac{K^2}{I} + 2h.$$

Multiplying this inequality by the positive function $-2\dot{I}$ and integrating the result over the interval (t_1, t) , where $t < t_0$, we get

$$\dot{I}^2(t_1) - \dot{I}^2(t) \geq 2K^2 \ln \frac{I(t_1)}{I(t)} + 4h(I(t_1) - I(t)).$$

This clearly gives the inequality

$$2K^2 \ln \frac{I(t_1)}{I(t)} \leq I^2(t_1) + 4|h|I(t_1),$$

from which we conclude that $I(t)$ has a positive lower bound in the interval (t_1, t_0) provided $K^2 \neq 0$. $\triangleright$

2.3. Binary Collisions. We say that a *binary collision* occurs at time t_0 if the distance between two points, say m_1 and m_n , tends to zero as $t \rightarrow t_0$, whereas the mutual distances between the remaining points are bounded from below by a positive constant for t close to t_0 . For such values of t the influence of the points $m_2, \dots, m_{n-1}$ upon the motion of m_1 and m_n is clearly negligible in comparison with the interaction between m_1 and m_n . It is therefore natural to expect that close to the collision time t_0 the vector $r_{1n}(t)$ will behave roughly as in the case of collisions in the two-body problem (§1). In that problem a local uniformizing variable was the true anomaly $u(t)$, proportional to the integral of the inverse of the distance between the points. Hence, in the case of a binary collision it is natural to attempt to regularize the solution by means of the variable

$$u(t) = \int_{t_0}^t \frac{ds}{|r_{1n}(s)|}. \quad (12)$$

One can show that this is indeed a good choice: the functions $r_k(u)$ are regular in a neighborhood of the point $u=0$ (corresponding to the binary collision); furthermore, $t(u) - t_0 = u^3 p(u)$, where $p(\cdot)$ is holomorphic in the neighborhood of $u=0$, and $p(0) \neq 0$. Therefore, in the case of a binary collision, as in the two-body problem, the coordinates of the points r_k are holomorphic functions of the variable $\sqrt[3]{t-t_0}$ and hence each admits a unique real analytic continuation for $t > t_0$. One can further show that functions $r_2(t), \dots, r_{n-1}(t)$ are actually holomorphic in a neighborhood of the point t_0 .

To ensure that the uniformizing variable $u(t)$ will be suitable for any pair of points and any time of binary collision, we must replace (12) by the formula

$$u(t) = \int_0^t V(s) ds = \int_0^t \sum_{j < k} \frac{m_j m_k}{|r_{jk}(s)|} ds.$$

If in the three-body problem the polar angular momentum is different from zero, then the only possible singularities are binary collisions. K.F. Sundman showed that the functions $r_k(u)$, $1 \leq k \leq 3$, are holomorphic in a strip $|\operatorname{Im} u| < \delta$ of the complex plane $u \in \mathbb{C}$ which contains the real axis. Now let us map this strip conformally onto the unit disc $|\omega| < 1$ by the transformation

$$\omega = (e^{\frac{\pi u}{2\delta}} - 1)(e^{\frac{\pi u}{2\delta}} + 1)^{-1},$$

which takes the real axis $-\infty < u < +\infty$ into the segment $-1 < \omega < 1$. Then the coordinates of the points r_k become functions holomorphic in the disc $|\omega| < 1$, and they can be expressed as convergent power series in the new variable ω . These series describe the motion of the three bodies for all values of the time t , from $-\infty$ through $+\infty$.¹⁷

This result was obtained by Sundman (1913). It followed earlier investigations of Poincaré and Weierstrass, in which convergent power series expansions in the auxiliary variable ω were obtained for the solutions of the n -body problem in the absence of collisions. As for collisions, in the three-body problem they are extremely rare. In fact, using the theorem on simultaneous collisions and also regularization of binary collisions one can show that in the 12-dimensional configuration space of the three-body problem (with the center of mass fixed) the collision trajectories lie on certain special analytic 10-dimensional surfaces. Their measure is, of course, zero. However, it is not known whether these surfaces may densely fill whole domains in the state space.

In conclusion we give, for illustrative purposes, the results of numerical calculations for the "Pythagorean" variant of the three-body problem, in which three bodies of masses 3, 4, and 5 are at rest at the initial time at the point of the (x, y) -plane with coordinates (1, 3), (-2, -1), and (1, -1), respectively.

Calculations of the Pythagorean three-body problem were initiated by C. Burrau as early as 1913 and have been continued in our time by V. Szebehely with the aid of fast electronic computers. In Figures 12-14 one can

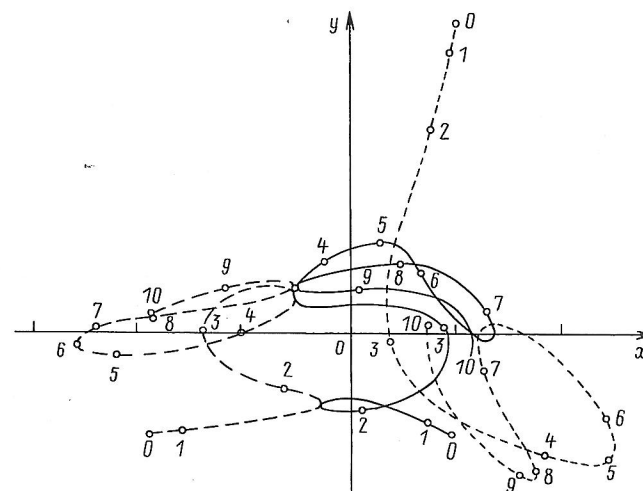


Fig. 12. Motion of the mutually attracting masses in the Pythagorean three-body problem over the time interval $0 \leq t \leq 10$

¹⁷ These power series in ω are however of no use in practical calculations because of their extremely low rate of convergence.

respectively see the points moving very close (near encounters of the points), binary collisions, and the dispersal of the triple system. Figure 15 shows a "final" motion: the point of mass $m=5$ moves along a straight line away from the "double star" constituted by the points $m=3$ and $m=4$, which collide periodically. It is worth noting that although in this case the angular momentum is zero, no triple collisions occur.

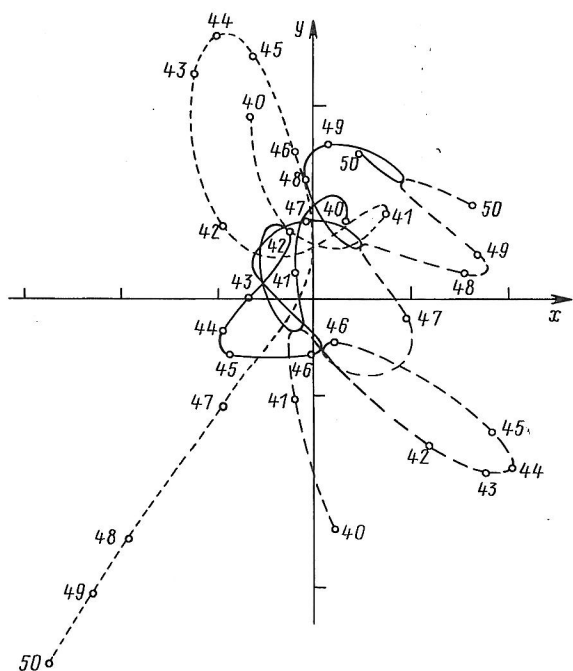


Fig. 13. Aspect of orbits in the Pythagorean three-body problem over the time interval $40 \leq t \leq 50$

2.4. Singularities of Solutions in the n -Body Problem. In the case of multiple collisions, that is, when $k \geq 3$ point masses interacting gravitationally collide simultaneously, the singularities of the coordinates r_k of the masses as functions of time have a rather complicated analytic structure. Generally, these singularities are not algebraic; moreover, the functions r_j ($1 \leq j \leq n$) do not admit a real analytic continuation beyond the collision time.

This can be seen even in examples of simultaneous collisions in the three-body problem. It turns out that for arbitrary values of the masses m_1, m_2, m_3 there exist solutions of the form

$$r_j(t) = t^{2/3} \sum_{m=0}^{\infty} a_{jm} t^{\alpha m}. \quad (13)$$

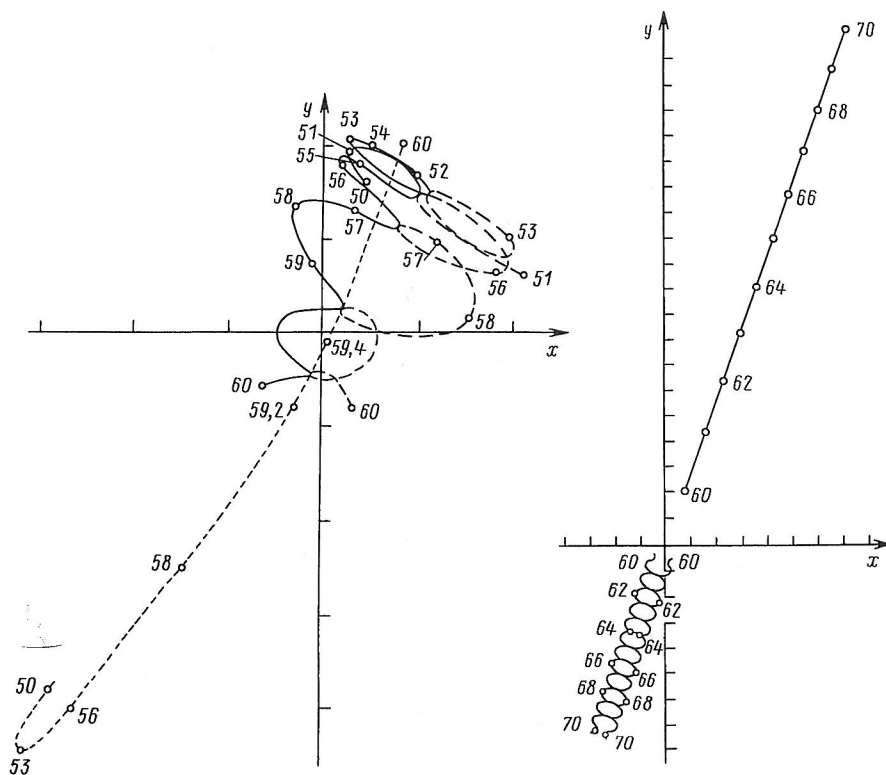


Fig. 14

Fig. 15

Fig. 14. Evolution of orbits in the Pythagorean three-body problem over the time interval $50 \leq t \leq 60$

Fig. 15. Formation of a double star in the Pythagorean three-body problem ($60 \leq t \leq 70$)

Here the positive number α is a nonconstant algebraic function of the masses m_1, m_2, m_3 , and the coefficients a_{jm} are not simultaneously zero. At time $t=0$ a triple collision occurs. In the typical case, where α is irrational, series (13) has an isolated logarithmic singularity at $t=0$. In particular, this solution, which is real for $t > 0$, has infinitely many distinct analytic branches for $t < 0$, which, however, are all complex.

Solution (13) was discovered by H. Block (1909) and J.F. Chazy (1918) by means of the following device: for arbitrary values of the masses, the equations of the three-body problem admit a "homographic" solution, for which the triangle formed by the bodies always remains similar to itself. This solution is analytically given by the formula

$$r_j(t) = a_{j0} t^{3/2} \quad (1 \leq j \leq 3). \quad (14)$$

Among the characteristic numbers of the variational equation (i.e., of the system of equations linearized about this solution), there is a negative number $-\alpha$. According to well-known results of Lyapunov and Poincaré, the equations of motion have a solution (13) which is asymptotic to solution (14). We remark that the method of Block and Chazy has been in fact used earlier (in 1894) by Lyapunov to prove that the solutions of the equations describing the rotation of a heavy rigid body around a fixed point are not single-valued as functions of complex time.

Consider a solution $r_k(t)$ of the three-body problem. Suppose that at the initial time t_0 all $r_{jk} \neq 0$ ($j \neq k$). Let us track this solution for $t > t_0$. Three cases are possible:

(a) there are no collisions for any $t > t_0$; then the motion proceeds without singularities till $t = +\infty$;

(b) at some instant $t_1 > t_0$ a "continuable" collision, i.e., one admitting analytic continuation, occurs;

(c) at some instant a "non-continuable" singularity, i.e., one for which analytic continuation is not possible, occurs.

Consider case (b). Then for $t > t_1$ one of the three variants (a)–(c) is again possible. Continuing this process, we may either reach case (a) or (c) after a finite number of steps, or observe infinitely many continuable collisions at times $t_1, t_2, \dots, t_k, \dots$. One can show that for $n=3$ in the latter case

$$\lim_{k \rightarrow \infty} t_k = +\infty.$$

In the n -body problem with $n \geq 4$, however, a principally different type of singularity is possible. Already in the 4-body problem on a line there exist motions in which infinitely many binary collisions occur over a finite time interval $[0, t_0]$. Moreover, for $t \rightarrow t_0$ three of the bodies diverge at infinity: one of them in a certain direction and the other two in the opposite direction, as in the Pythagorean three-body problem. However, in contrast to the three-body problem, the colliding bodies unboundedly approach one another, and this provides the kinetic energy which causes the solution to become unbounded in finite time. The fourth body bounces back and forth between them. When the fourth body passes close to the two bodies which are approaching one another, then some kind of an almost triple collision occurs. The existence of such a motion was demonstrated by J. Mather using the McGehee regularization of the simultaneous collision in the three-body problem (see [136]).

§3. Particular Solutions

In the n -body problem only a few exact solutions have been found. Basically, all these solutions were already known to Euler and Lagrange.

3.1. Central Configurations. We shall say that in an barycentric reference frame n point masses (m_i, r_i) form a *central configuration* if

$$\frac{\partial V}{\partial r_i} = \sigma \frac{\partial I}{\partial r_i}, \quad 1 \leq i \leq n, \quad (15)$$

where

$$V = \sum_{j < k} \frac{m_k m_j}{|r_{kj}|}$$

is the gravitational force function $I = \sum m_i r_i^2$ is the polar moment of inertia, and the scalar function σ does not depend on the index i . It follows from Euler's formula for homogeneous functions that $\sigma = -V/2I$. Thus, we can

$$I \frac{\partial V}{\partial r_i} = -\frac{1}{2} V \frac{\partial I}{\partial r_i}.$$

That is to say, to central configurations there correspond critical points of the function IV^2 . Since this function is homogeneous, it follows that if (m_i, r_i) is a central configuration, then so is $(m_i, \alpha r_i)$ for every $\alpha \neq 0$; we shall not distinguish between such central configurations.

Finding all central configurations for an arbitrary number n of points is a difficult problem, which is still open. Leaving aside the trivial case $n=2$, we list the result known presently in this direction.

For $n=3$ the only noncollinear central configuration is formed by the vertices of an equilateral triangle (Lagrange). For $n=4$ the unique noncoplanar (nonflat) configuration is given by the vertices of a regular tetrahedron. The collinear central configurations are described by the following *theorem of Moulton* [42]: each enumeration of the points uniquely determines a central configuration in which the points lie collinearly in the given order. Therefore, there are exactly $n!/2$ distinct collinear central configurations. For $n=3$ there are exactly three such configurations, discovered by Euler.

The notion of central configuration proves useful in the analysis of simultaneous collisions: it turns out that the configuration of gravitating points at the instant of a simultaneous collision is (in an asymptotic sense) central. From formula (15) it follows that if at the initial time the points form a central configuration and are at rest, then their configuration obviously does not change until the simultaneous collision occurs.

3.2. Homographic Solutions. A given solution of the n -body problem is called *homographic* if in an inertial barycentric reference frame the configuration formed by the n bodies remains similar to itself at all moments of time. If, in this case, the configuration does not rotate, then the solution will be referred to as *homothetic*. As an example we offer the configurations mentioned at the end of §3.1. If, however, the configuration remains congruent to itself (in the sense of Euclidean geometry, i.e., it rotates without dilation) then the solution is called a *relative equilibrium*.

It is not hard to show that

a) an homographic solution is homothetic if and only if the polar angular momentum is equal to zero;

b) an homographic solution is a relative equilibrium if and only if it is planar and the corresponding configuration rotates with constant nonzero angular velocity.

The next two facts are less readily established:

c) if an homographic solution is not coplanar, then it is homothetic;

d) if an homographic solution is coplanar, then it is planar.

In particular, every homographic solution is either planar or homothetic. In the three-body problem, for every homographic solution, the three bodies in a barycentric reference frame remain in an invariable plane which contains the center of mass (Lagrange).

Proposition 4. *In an homographic solution the bodies form a central configuration at any moment of time.*

This result provides a method for constructing homographic solutions. As an example we give here the well-known *theorem of Lagrange* (1772).

Theorem 4. *For arbitrary values of the masses the three-body problem admits an exact solution with the following properties:*

1) *the plane which contains the point masses is fixed in a barycentric reference frame;*

2) *the resultant of the Newtonian forces of attraction applied to each of the three masses passes through the common barycenter;*

3) *the triangle formed by the three bodies is equilateral;*

4) *the trajectories of the bodies are similar conics with one focus at the common barycenter.*

In the particular case of equal masses these conics are congruent and differ from each other by a rotation through 120° . This observation is readily generalized: the problem of n bodies of equal masses has solutions in which each body describes a conic with a focus at the common center of mass; their trajectories are congruent and differ one from another by a rotation through the angle $2\pi/n$.

3.3. The Amended Potential and Relative Equilibria

Proposition 5. *The relative equilibria configurations with polar angular momentum K coincide with the critical points of the function*

$$U_K = U + \frac{K^2}{2I}.$$

The function U_K is called the *amended* (reduced, or effective) potential. It was already used in §1, Ch. 1, to describe the regions of possible motions

in the planar n -body problem, and also in §2.1 for determining the trajectories in the two-body problem.

◁ Suppose that a relative equilibrium configuration rotates around the center of mass with constant angular velocity ω . Then clearly $K = I\omega$. Let us pass to a reference frame with coordinates u, v , which rotates with angular velocity ω . In the new frame the given relative equilibrium configuration is fixed. The Lagrangian here is

$$L = T + V = \frac{1}{2} \sum m_i (\dot{u}_i^2 + \dot{v}_i^2) + \omega \sum m_i (u_i \dot{v}_i - \dot{u}_i v_i) + V_\omega,$$

where $V_\omega = V + I\omega^2/2$. The corresponding equations of motion are

$$\begin{aligned} m_i \ddot{u}_i &= 2m_i \omega \dot{v}_i + \frac{\partial V_\omega}{\partial u_i}, \\ m_i \ddot{v}_i &= -2m_i \omega \dot{u}_i + \frac{\partial V_\omega}{\partial v_i}. \end{aligned} \quad (16)$$

Proposition 2 follows easily from these equations upon observing that the functions U_K and V_ω have the same critical points because at these points $K = I\omega$. ▷

§4. Final Motions in the Three-Body Problem

4.1. Classification of Final Motions According to Chazy

Theorem 5 (Chazy, 1922). *Every solution $r_k(t)$, $k=1, 2, 3$, of the three-body problem belongs to one of the following seven classes:*

1°. H (hyperbolic motions): $|r_k| \rightarrow \infty$, $|\dot{r}_k| \rightarrow c_k > 0$ as $t \rightarrow +\infty$;

2°. HP_k (hyperbolic-parabolic motions): $|r_i| \rightarrow \infty$, $|\dot{r}_k| \rightarrow 0$, $|\dot{r}_i| \rightarrow c_i > 0$ ($i \neq k$);

3°. HE_k (hyperbolic-elliptic motions): $|r_i| \rightarrow \infty$, $|\dot{r}_i| \rightarrow c_i > 0$ ($i \neq k$), $\sup_{t \geq t_0} |r_k| < \infty$;

4°. PE_k (parabolic-elliptic motions): $|r_i| \rightarrow \infty$, $|\dot{r}_i| \rightarrow 0$ ($i \neq k$), $\sup_{t \geq t_0} |r_k| < \infty$.

5°. P (parabolic motions): $|r_i| \rightarrow \infty$, $|\dot{r}_i| \rightarrow 0$.

6°. B (bounded motions): $\sup_{t \geq t_0} |r_k| < \infty$.

7°. OS (oscillatory motions): $\lim_{t \rightarrow +\infty} \sup_k |r_k| = \infty$, $\lim_{t \rightarrow +\infty} |r_k| < \infty$.

Examples of motions of the first six types were known to Chazy. The existence of oscillatory motions was proved by K.A. Sitnikov in 1959.

It is only natural to attach to these seven types of final motions subsets of the 12-dimensional phase space M^{12} of the three-body problem with fixed center of mass: each such subset consists of all the phase trajectories that correspond to motions of a given type. The qualitative picture of the partition

of M^{12} into classes of final motions is shown in Fig. 16. The sets H and HP_k lie in the domain in which the total energy h is positive, P lies on the hypersurface $h=0$, and the sets B , PE_k , and OS in the domain $h<0$; notions of class HE_k are possible for all values of h . It is known that H and HE_k are open in M^{12} , that HP_k consist of analytic manifolds of codimension 1, and that P consists of three connected manifolds of codimension 2 (represented by the three points in Fig. 16) and a manifold of codimension 3 (which is not shown in Fig. 16). The topology of the other classes of motions is not sufficiently studied.

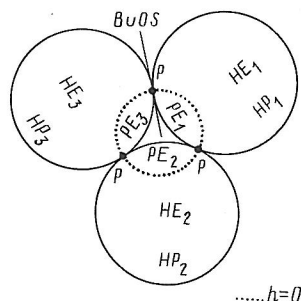


Fig. 16

4.2. Symmetry of Past and Future. By a theorem of Chazy one can introduce seven analogous *final classes* of motions when t tends to $-\infty$ instead of $+\infty$. To distinguish the classes corresponding to the cases $t \rightarrow \pm\infty$ we shall use superscripts (+) and (-) to write H^+ , HE_3^- , and so on. In one of his works (1929) Chazy mistakenly asserted that for each given solution of the three-body problem the final types of behavior for $t \rightarrow +\infty$ and $t \rightarrow -\infty$ coincide. There has been a longheld belief in this "symmetry" of past and future, despite a numerical counterexample constructed by L. Becker (1920) which indicated the possibility of "exchange": $HE_1^- \cap HE_2^+ \neq \emptyset$. Becker's example was attributed to errors in the numerical integration. In 1947 O.Yu. Schmidt indicated an example of "capture" in the three-body problem: $H^- \cap HE^+ \neq \emptyset$. This example, also constructed by numerical calculations, was proposed by Schmidt to support his well-known cosmological hypothesis. A rigorous proof of the possibility of capture was offered by K.A. Sitnikov in 1953.

The present state of our knowledge of final motions in the three-body problem is sketchily shown in Tables 1 and 2, which are taken from the work of V.M. Alekseev [3]. To each cell there corresponds one of the logically possible combinations of final types of past and future motion. The Lebesgue measure of the corresponding subsets of M^{12} is indicated (where it is known).

Table 1

		$t \rightarrow +\infty$	
		H^+	HE_i^+
$t \rightarrow -\infty$	H^-	Lagrange 1772 (isolated examples); Chazy 1922 Measure >0	PARTIAL CAPTURE Measure >0 Schmidt (numerical example), 1947 Sitnikov (qualitative methods), 1953
	HE_j^-	COMPLETE DISPERSAL Measure >0	$i=j$ Measure >0 Birkhoff 1927 $i \neq j$ EXCHANGE Measure >0 Becker (numerical examples), 1920 Alekseev (qualitative methods), 1956

Table 2

		$t \rightarrow +\infty$		
		HE_i^+	B^+	OS^+
$t \rightarrow -\infty$	HE_j^-	$i=j$ Measure >0 Birkhoff, 1927 EXCHANGE $i \neq j$ Measure >0 Becker, 1920 (numerical examples) Alekseev, 1956 (qualitative methods)	COMPLETE CAPTURE $\begin{cases} \text{Measure} = 0 \\ \text{Chazy, 1920} \\ \text{Merman, 1954} \end{cases}$ Littlewood, 1952 Alekseev, 1968 $\neq \emptyset$	$\begin{cases} \text{Measure} = 0 \\ \text{Chazy, 1929} \\ \text{Merman, 1954} \end{cases}$ Alekseev, 1968 $\neq \emptyset$
	B^-	PARTIAL DISPERSAL $\neq \emptyset$ Measure $=0$	Euler, 1772 Lagrange, 1772 Poincaré, 1892 (isolated examples); Measure >0 Arnold, 1963	Littlewood, 1952 Measure $=0$; Alekseev, 1968 $\neq \emptyset$
	OS^-	Measure $=0, \neq \emptyset$	Measure $=0, \neq \emptyset$	Sitnikov, 1959 $\neq \emptyset$ Measure $=?$

§5. The Restricted Three-Body Problem

5.1. Equations of Motion. The Jacobi Integral. Suppose that the Sun S and Jupiter J revolve around their center of mass in circular orbits. We choose units of length, time and mass so that the angular velocity of rotation,

the sum of masses of S and J , and the gravitational constant are all equal to one. It is readily verified that for this choice the distance between S and J is also equal to one.

Consider the motion of an asteroid A in the plane of the orbits of S and J . We assume that the mass of A is considerably smaller than the masses of the Sun and Jupiter, and accordingly we neglect the influence of A on the motion of the two larger bodies.

It is convenient to pass to a moving reference frame which rotates with unit angular velocity around the center of mass of the S, J system; in this frame S and J are at rest. We choose coordinates x, y in the moving frame so that the points S and J lie invariably on the x -axis and their center of mass is the origin of the coordinate system. Then the equations governing the motion of the asteroid can be written in the form (see (16)):

$$\begin{aligned}\ddot{x} &= 2\dot{y} + \frac{\partial V}{\partial x}, & \ddot{y} &= -2\dot{x} + \frac{\partial V}{\partial y}, \\ V &= \frac{x^2 + y^2}{2} + \frac{1-\mu}{\rho_1} + \frac{\mu}{\rho_2},\end{aligned}\quad (17)$$

where μ is Jupiter's mass and ρ_1, ρ_2 are the distances from the asteroid A to S and J . Since S and J have coordinates $(-\mu, 0)$ and respectively $(1-\mu, 0)$,

$$\rho_1^2 = (x + \mu)^2 + y^2, \quad \rho_2^2 = (x - 1 + \mu)^2 + y^2.$$

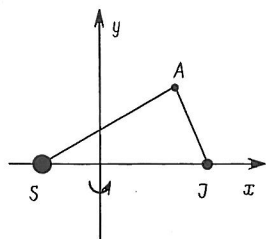


Fig. 17. The restricted three-body problem

Equations (17) have first integral

$$\frac{\dot{x}^2 + \dot{y}^2}{2} - V(x, y) = h,$$

known as the *Jacobi integral*. It expresses the conservation of energy in the relative motion of the asteroid.

For fixed h the motion of the asteroid takes place in the *Hill region*

$$\{(x, y) \in R^2: V(x, y) + h \geq 0\}.$$

5.2. Relative Equilibria and the Hill Region. The shape of the Hill region depends on the position of the critical points of function $V(x, y)$. To each such critical point (x_0, y_0) there corresponds an "equilibrium" solution $x(t) \equiv x_0, y(t) \equiv y_0$, which may naturally be termed a relative equilibrium. Let us show that for each value $\mu \in (0, 1)$ there are exactly five such points.

We calculate

$$\begin{aligned}V'_y &= yf, & f &= 1 - \frac{1-\mu}{\rho_1^3} - \frac{\mu}{\rho_2^3}, \\ V'_x &= xf - \mu(1-\mu)\left(\frac{1}{\rho_1^3} - \frac{1}{\rho_2^3}\right),\end{aligned}$$

and now solve the system of algebraic equations $V'_x = V'_y = 0$. Suppose first that $y \neq 0$. Then $f = 0$, hence $\rho_1 = \rho_2 = \rho$. From the equation $f = 0$ we find that $\rho = 1$. Thus, in this case the points S, J , and A are the vertices of an equilateral triangle. There are exactly two such equilibria, called *equilateral* or *triangular libration points*. They should be thought of as a particular case of Lagrange's solutions of the general, "unrestricted" three-body problem (see §3). Lagrange himself regarded these solutions as a "pure curiosity" and considered them to be useless in astronomy. However, 1907 brought the discovery of an asteroid, Achilles, which moves practically on Jupiter's orbit being constantly ahead of it by 60° . Close to Achilles there are nine other asteroids (the "Greeks"), and five other (the "Trojans") were discovered on the opposite side, which also form an equilateral triangle with the Sun and Jupiter.

Now let us turn to the relative equilibria lying on the x -axis. They are the critical points of the function

$$g(x) = \frac{x^2}{2} + \frac{1-\mu}{|x+\mu|} + \frac{\mu}{|x-1+\mu|}.$$

Since $g(x) > 0$ and $g(x) \rightarrow +\infty$ as $x \rightarrow \pm\infty, x \rightarrow -\mu$, and $x \rightarrow 1-\mu$, it follows that in the intervals $(-\infty, -\mu), (-\mu, 1-\mu), (1-\mu, +\infty)$ into which the points S and J divide the x -axis there exist three local minima of g . Since $g''(x) > 0$, these are the only critical points of g . They were found by Euler and are called *collinear libration points*.

One can show that the collinear libration points (we denote them by L_1, L_2, L_3) are of hyperbolic type, whereas the equilateral libration points (L_4 and L_5) are nondegenerate minima of function V . Figure 18 shows how the topology of Hill's region changes as the Jacobi constant h varies from $-\infty$ to $+\infty$; the mass of Jupiter is taken to be smaller than that of the Sun. If h is larger than the negative number

$$-\frac{1}{2}(3 - \mu + \mu^2),$$

then Hill's region coincides with the entire plane $R^2 = \{(x, y)\}$. For $\mu = 1/2$ the Hill region is symmetric not only about the x -axis, but also about the y -axis.



Fig. 18

The collinear libration points are always unstable: among the roots of the characteristic equation of the variational equations (i.e., the linearized equations of motion) there are some having positive real parts. For the case of equilateral libration points these roots are purely imaginary and distinct only when

$$27\mu(1-\mu) < 1. \quad (18)$$

If this condition is fulfilled, then the equilateral relative equilibria are stable in the first approximation. The investigation of the Lyapunov stability of these points turns out to be a considerably more difficult task; we postpone it until Chapter 7. We remark in conclusion that for the real Sun-Jupiter system condition (18) is clearly fulfilled.

5.3. Hill's Problem. Let us take as the origin of the rotating coordinate system the point where the body of the mass μ lies. Then the coordinates x, y of the third, small-mass body must be replaced by $x-(1-\mu), y$. Preserving the notation x, y for these new variables, we see that the equations of motion have the same form (17), with the only difference being that the potential is now the function

$$V = (1-\mu)x + \frac{1}{2}(x^2 + y^2) + (1-\mu)(1+2x+x^2+y^2)^{-1/2} + \mu(x^2+y^2)^{-1/2}. \quad (19)$$

We follow G.W. Hill and introduce a further simplification of the problem, borrowed from astronomy. Suppose again that the body of mass $1-\mu$ represents the Sun, μ is the Earth, and that a third body, the Moon, of negligible mass, moves in the vicinity of the point $(0, 0)$ where the Earth is at rest. Let us neglect in equations (17) all the terms of order two or higher in x, y . Equivalently, we ignore in (19) the terms of order at least three in x, y . Modulo such terms, function V is given by

$$V = \frac{\mu}{2}(x^2 + y^2) + \frac{3}{2}(1-\mu)x^2 + \mu(x^2 + y^2)^{-1/2}.$$

Since the Earth's mass μ is considerably smaller than the Sun's mass $1-\mu$, we can also neglect the first term in this expression of V .

It is convenient to change the units of mass and length slightly as follows:

$$x \rightarrow \alpha x, \quad y \rightarrow \alpha y, \quad \mu \rightarrow \beta \mu, \quad 1-\mu \rightarrow \beta(1-\mu),$$

where

$$\alpha = \left(\frac{\mu}{1-\mu} \right)^{1/3}, \quad \beta = (1-\mu)^{-1}.$$

Following these transformations, the equations governing the motion of the Moon take the form

$$\ddot{x} - 2\dot{y} = \frac{\partial V}{\partial x}, \quad \ddot{y} + 2\dot{x} = \frac{\partial V}{\partial y}, \quad V = \frac{3}{2}x^2 + (x^2 + y^2)^{-1/2}. \quad (20)$$

They possess the Jacobi first integral

$$\frac{1}{2}(\dot{x}^2 + \dot{y}^2) - V(x, y) = h.$$

It is readily verified that on passing from the restricted three-body problem to its limit version just described, known as the *Hill problem*, two of the equilateral libration points and one of the collinear ones disappear. In fact, the system of equations $V'_x = V'_y = 0$ has only the two solutions $(x, y) = (\pm 3^{-1/3}, 0)$. Hill's region

$$\{V(x, y) + h \geq 0\}$$

is symmetric relative to the x - and y -axes for every value of h . If $h \geq 0$, Hill's region coincides with the entire plane. For $h < 0$ its boundary has two asymptotes parallel to the y -axis: $x = \pm(-2/3h)^{1/2}$. The shape of these regions depends on whether the constant $(-h)$ is larger than, equal to, or smaller than the unique critical value of V , which is $3^{1/3} \cdot 3/2$. These three cases are depicted in Fig. 19. The only case of interest in astronomical applications is (a), and there only a domain surrounding the origin of coordinates.

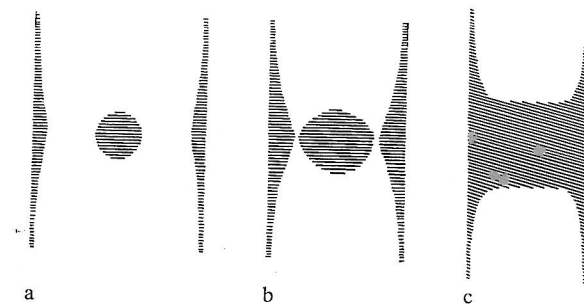


Fig. 19

We now turn to various questions connected with the regularization of Hill's problem. To this end, we shift to new, parabolic coordinates through the formulas $x = \xi^2 - \eta^2$, $y = 2\xi\eta$, and also change the time variable t along the trajectories to τ according to the rule

$$\frac{dt}{d\tau} = 4(\xi^2 + \eta^2).$$

Denoting differentiation with respect to τ by a prime, we write the equations of motion in the new variables as

$$\xi'' - 8(\xi^2 + \eta^2)\eta' = \hat{V}'_\xi, \quad \eta'' + 8(\xi^2 + \eta^2)\xi' = \hat{V}'_\eta,$$

here

$$\hat{V} = 4 + 4(\xi^2 + \eta^2)h + 6(\xi^2 - \eta^2)(\xi^2 + \eta^2).$$

The energy integral takes the form

$$\frac{1}{2}(\xi'^2 + \eta'^2) - \hat{V}(\xi, \eta, h) = 0.$$

This regularization of Hill's problem, proposed by Birkhoff, permits a straightforward investigation of the analytic singularities of the solutions corresponding to Moon-Earth collisions. Suppose that such a collision occurs at the time $t = 0$ and let $\tau(0) = 0$. Then obviously

$$\xi = (\sqrt{8} \sin \alpha) \tau + \dots, \quad \eta = (\sqrt{8} \cos \alpha) \tau + \dots, \quad \text{and} \quad t = \frac{3}{2} \tau^3 + \dots,$$

where α is an integration constant. Thus, the new time τ is a uniformizing variable and, as in the case of binary collisions in the general three-body problem, the solution $\xi(t), \eta(t)$ admits a unique real analytic continuation beyond the collision time.

As we have already mentioned, in astronomy one is interested only in notions occurring near the point $\xi = \eta = 0$. For negative h of large absolute value it is convenient to use the new variables

$$\varphi = 2\xi \{-2h - 3(\xi^2 - \eta^2)^2\}^{1/2}, \quad \psi = 2\eta \{-2h - 3(\xi^2 - \eta^2)^2\}^{1/2}.$$

Then the energy integral takes the very simple form

$$\xi'^2 + \eta'^2 + \varphi^2 + \psi^2 = 8.$$

This is the equation of a three-dimensional sphere in the four-dimensional phase space of variables $\xi', \eta', \varphi, \psi$. Since to the points (ξ, η) and $(-\xi, -\eta)$ here corresponds one and the same point in the (x, y) -plane, Moon's states $(\xi', \eta', \varphi, \psi)$ and $(-\xi', -\eta', -\varphi, -\psi)$ must be identified. We thus conclude that for negative h of large absolute value the component of the three-dimensional energy level which is of interest to us is diffeomorphic to the three-dimensional projective space. This is of course valid for all $h < -\frac{3}{2} 3^{1/3}$.

We conclude this section by a discussion of certain periodic solutions of Hill's problem, which have important astronomical applications. Specifically, we are interested in periodic solutions $x(t), y(t)$ of small period τ , near the Earth (the point $x=y=0$), whose orbits are symmetric relative to the x - and y -axes. These symmetry requirements are expressed by the equalities

$$x(-t) = x(t) = -x\left(t + \frac{\tau}{2}\right), \quad y(-t) = -y(t) = y\left(t + \frac{\tau}{2}\right).$$

Accordingly, we seek these solutions as trigonometric series

$$x(t) = \sum_{n=-\infty}^{\infty} a_n(m) \cos(2n+1) \frac{t}{m}, \quad y(t) = \sum_{n=-\infty}^{\infty} a_n(m) \sin(2n+1) \frac{t}{m},$$

where

$$m = \frac{\tau}{2\pi}.$$

Substituting these series in the equations of motion (20) we obtain an infinite nonlinear system of algebraic equations for infinitely many unknown coefficients $a_n(m)$. Hill showed (1878) that this system admits a unique solution, at least for small values of m (see [36, 42]). The value $m_0 = 0.08084\dots$ for the real Moon falls in this admissible range. The convergence of Hill's series was established by Lyapunov in 1895.

One can show that the following asymptotic expansions hold for the coefficients $a_n(m)$:

$$a_0 = m^{2/3} \left(1 - \frac{2}{3}m + \frac{7}{18}m^2 - \dots\right),$$

$$\frac{a_1}{a_0} = \frac{3}{16}m^2 + \frac{1}{2}m^3 + \dots, \quad \frac{a_{-1}}{a_0} = -\frac{19}{16}m^2 - \frac{5}{3}m^3 - \dots,$$

$$\frac{a_2}{a_0} = \frac{25}{256}m^4 + \dots, \quad \frac{a_{-2}}{a_0} = 0 \cdot m^4 + \dots, \dots$$

This shows that for small values of m the main contribution in Hill's periodic solutions comes from the terms

$$x_0(t) = m^{2/3} \cos \frac{t}{m}, \quad y_0(t) = m^{2/3} \sin \frac{t}{m},$$

which represent the law governing the motion of the Moon when the influence of the Sun is neglected. The presence of the coefficient $m^{2/3}$ is a consequence of Kepler's third law. For small values of the parameter m the Sun, while perturbing the Earth-Moon system, does not destroy the circular periodic orbits, merely deforming them slightly.

§ 6. Ergodic Theorems in Celestial Mechanics

6.1. Stability in the Sense of Poisson. Let (M, S, μ) be a complete measure space; here S is a σ -algebra of subsets of M , and μ is a countably-additive measure on S . Consider a measure-preserving automorphism g of M . We call the sets

$$\Gamma_p = \bigcup_{n \in \mathbb{Z}} g^n(p), \quad \text{where } g^0(p) = p,$$

and

$$\Gamma_p^+ = \bigcup_{n \geq 0} g^n(p)$$

the *trajectory* and the *positive* (or *forward*) *semitrajectory*, respectively, of the point $p \in M$.

Poincaré's Recurrence Theorem. Let $\mu(M) < \infty$. Then for every measurable set $V \in \mathcal{S}$ of positive measure there is a set $W \subset V$ with $\mu(W) = \mu(V)$ such that for every $p \in W$ the set $\Gamma_p^+ \cap W$ is infinite.

We follow Poincaré and apply this result to the restricted three-body problem. In the notations of the preceding section, the equations governing the motion of the asteroid are Lagrange's equations with the Lagrangian

$$L = \frac{1}{2}(\dot{x}^2 + \dot{y}^2) + x\dot{y} - y\dot{x} + V, \quad V = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{\rho_1} + \frac{\mu}{\rho_2}.$$

These can be written as Hamilton's equations with the Hamiltonian

$$H = \frac{1}{2}(X^2 + Y^2) + yX - xY - G, \quad G = \frac{\mu}{\rho_1} + \frac{1-\mu}{\rho_2},$$

where $X = \dot{x} - y$ and $Y = \dot{y} + x$ are the momenta canonically conjugate to the coordinates x and y . According to Liouville's theorem, the phase flow of this system, g^t , preserves the usual Lebesgue measure in $R^4 = \{(X, Y, x, y)\}$.

Consider the set of all phase points at which $c_1 < -H < c_2$, where c_1 and c_2 are sufficiently large positive constants. As we have seen in §5, if this double inequality holds, then the point (x, y) lies in one of the three connected subregions of the Hill region $\{G \leq c_1\}$. Pick one of the two regions containing the Sun and Jupiter. The corresponding connected region in phase space is obviously invariant under the action of g^t . From it we must eliminate the collision trajectories; the union of these trajectories has measure zero. Denoting the resulting set by M , we show that its measure is finite. In fact, the coordinates (x, y) of the points of M belong to a bounded subset of the plane $R^2 = \{(x, y)\}$. The admissible momenta X and Y are subject to the inequalities

$$2(G - c_2) < (Y + x)^2 + (Y - y)^2 < 2(G - c_1),$$

which are consequences of the Jacobi integral. These inequalities determine a circular annulus in the plane $R^2 = \{(X, Y)\}$ whose area does not exceed $2\pi(c_2 - c_1)$. It follows from these remarks that the measure $\mu(M)$ is finite, and hence that Poincaré's recurrence theorem is applicable: for almost every $p \in M$ the semitrajectory $\{g^t(p), t \geq 0\}$, intersects any neighborhood of the point p for arbitrarily large values of t . Poincaré called the motions with this property *stable in the sense of Poisson* (or *Poisson-stable*).

6.2. Probability of Capture. Again, let V be a measurable set of positive measure. For $n \in \mathbb{N}$ we let V^n denote the set of points $p \in V$ such that $g^k(p) \in V$ for all $0 \leq k \leq n$. Clearly, $V^{n_1} \supset V^{n_2}$ whenever $n_1 < n_2$. Set

$$B = \bigcap_{n \geq 0} V^n.$$

The set B is measurable and $\mu(B) < \infty$. If $p \in B$, then obviously $\Gamma_p^+ \subset V$.

Put $B^n = g^n(B)$. The B^n are measurable, and again $B^{n_1} \supset B^{n_2}$ if $n_1 < n_2$. The set

$$D = \bigcap_{n \geq 0} B^n \subset B$$

is also measurable, and obviously $\Gamma_p \subset V$ for all $p \in D$.

Proposition 6. $\mu(B \setminus D) = 0$.

In order that this assertion be useful we need first to verify that $\mu(B) > 0$. However, in concrete problems proving this might in fact be very difficult. Proposition 6, which is attributed to K. Schwarzschild, is of course valid also for the case of continuous time n .

Suppose, for example, that the Sun-Jupiter system has "captured" from the surrounding space asteroids (the "Greeks" and the "Trojans") in the vicinity of the equilateral libration points. Proposition 6 immediately shows that this event has probability zero. Thus, "capture" phenomena in celestial mechanics should be considered only in mathematical models which account for dissipation of energy.

A more interesting example is provided by the following reasoning of J. Littlewood. Consider the n -body problem with the center of mass at rest. The motion of the n points is described by a Hamiltonian system, whose Hamiltonian is regular in the domain in which the mutual distances $|r_{kl}|$ are different from zero. For arbitrary $c > 1$ we consider the open subset $A(c)$ of phase space defined by the inequalities

$$c^{-1} < |r_{kl}| < c \quad (1 \leq k < l \leq n), \quad -c < H < c.$$

Since $A(c)$ is bounded, $\mu(A(c)) < \infty$. Therefore, by Proposition 6, the set $B(c)$ of points which remain in $A(c)$ for all $t \geq 0$ differs only by a set of measure zero from the set $D(c)$ of those points which remain in $A(c)$ for all $t \in \mathbb{R}$.

If $c_1 < c_2$ then obviously $A(c_1) \subset A(c_2)$, $B(c_1) \subset B(c_2)$, and $D(c_1) \subset D(c_2)$. Hence, a corresponding assertion holds true for the sets

$$A = \bigcup_{c > 1} A(c), \quad B = \bigcup_{c > 1} B(c), \quad D = \bigcup_{c > 1} D(c).$$

Thus, for the points $p \in B$ the mutual distances $|r_{kl}|$ remain bounded from above and from below for all $t \geq 0$ by some positive constants which do not depend on p . For points $p \in D$ this property holds for all $t \in \mathbb{R}$. Almost all points of D belong to B .

Suppose, for example, that it is known that the planetary system is stable "in the past". If it captures a new body, say, a speck of dust arriving from "infinity", then the newly formed system of bodies loses its stability: with probability one, either a collision occurs, or one of the bodies escapes again at infinity. Moreover, it is by no means necessary that the speck of dust is the one that leaves the Solar system: Jupiter, or even the Sun may equally well leave!